GDP calculation

Formulas:

Calculating GDP using expenditure approach:

$$Y=C+I+G+NX$$

Here,
C= Consumption
I=Investment, (it will include Business capital, Residential Capital and Inventory)
G= Government Expenditure, transfer payment needs to subtracted from it
NX=net export, Export-Import or, X-M

GDP deflator:

The GDP deflator is a measure of inflation in an economy, specifically focusing on the price changes of all goods and services produced within that economy. It is a price index that reflects how much the average price level of goods and services has changed over time. If the GDP deflator for a country increases from 100 to 110 between two years, it means that, on average, prices for all goods and services produced in that country increased by 10% over that period.

Calculating GDP Deflator

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \text{ X } 100$$

$$\text{Nominal GDP} = \frac{\text{(Deflator) x (Real GDP)}}{100}$$

$$Real\ GDP = \frac{Nominal\ GDP}{GDP\ Deflator}\ x\ 100$$

GDP Growth rate:

The economic growth rate can be measured as the annual percentage change of real GDP. The growth rate of real GDP equals:

$$\frac{\text{(Real GDP in current year - Real GDP in previous year)}}{\text{(Real GDP in previous year)}} \times 100$$

Example:

In 2022-23 fiscal year, a country's consumption was $1,000 billion, Investment was $500 billion, Government Spending was $450 billion, Transfer payment was $50 billion, Export was $300 billion, import was $100 billion. GDP deflator was 1.05. Calculate real GDP of 2022-23 FY.

Suppose, in FY 2023-24 real GDP was $2300 billion. Now calculate real gdp growth rate between FY 2022-23 and FY 2023-24.

Solution:

Here, in FY 2022-23,

Consumer Spending (C): $1,000 billion
Investment Spending (I): $500 billion
Government Spending (G): ($450- $50) billion= $400 billion
Exports (X): $300 billion
Imports (M): $100 billion
Net Export(NX): ($300-$100) billion= $200 billion

Calculating GDP in expenditure approach:

GDP (Y) = $1,000 billion + $500 billion + $400 billion + $200 billion
= $2,100 billion

This is the nominal GDP.

Here, GDP deflator is 1.05
So, real GDP = ($2,100 billion/1.05)
= $2000 billion

So real GDP growth rate from FY 2022-23 to FY 2023-24 will be
{($2300-$2000)/$2000}x100
=($300/$2000)x100
=15%

Ans: real GDP of 2022-23 FY is: $2000 billion
real gdp growth rate between FY 2022-23 and FY 2023-24 is 15%.